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Project Proposal

(SCOPE DOCUMENT)

for

NeuraPath

Version 1.0

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Scope Document Revision History

No.	Comment	Action

Supervisor Signature: _____

Date: _____

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Project Category: **(**Select all the major domains of proposed project**)**

○ **A-**Desktop Application/Information System	○ **B-**Web Application/Web Application Based Information System	
○ **C-**Problem Solving and Artificial Intelligence	○ **D-**Simulation and Modeling	● **E-**Smart Phone Application
○ **F-**Smart Phone Game	○ **G-**Networks	○ **H-**Image Processing
○ Other (Specify Category): _		

1 Abstract

NeuraPath (Neural + Path) AI-powered pathfinding for careers that leverages artificial intelligence to enhance career decision-making for students, job seekers, and employed professionals. The platform combines machine learning algorithms and natural language processing to deliver core functionalities: personalized career recommendations for students, AI-powered resume optimization for job seekers, and career progression guidance for working professionals.

For students, the platform analyzes academic profiles, skills, and interests to suggest suitable career paths, additionally guiding them towards relevant job opportunities within their chosen domain or field. Job seekers benefit from automated resume optimization, including ATS compliance scoring, keyword optimization, and formatting improvements to maximize job opportunities. For employed professionals, the system analyzes their current position, salary, and skills to map out promotion pathways against industry standards. By bridging the gap between education and employment, this platform aims to reduce career uncertainty and enhance employability across all user segments.

2 Introduction

This project proposes the development of NeuraPath, a comprehensive AI- powered platform aimed at revolutionizing career guidance for a diverse user base, including students, aspiring job seekers, and employed professionals. Leveraging cutting-edge machine learning and natural language processing, the system provides personalized career recommendations

For **students**, the platform offers Personalized career recommendations based on ML models trained on industry data and Skill gap analysis to identify missing competencies for desired roles.

For **Job seekers**, it offers ATS compliance checks and keyword enhancement using real-time job market data. It identifies skill gaps and improves resume design for professional appeal. The system matches profiles with relevant job openings. Powered by NLP and web scraping, it ensures resumes stand out to employers.

For **employed professionals**, NeuraPath offers promotion road-maps and skill gap analysis to guide career advancement. By integrating real-time labor market data from platforms like LinkedIn, the solution ensures recommendations align with current industry demands.

3 Objective

The **Objective** of NeuraPath is to guide career decisions for students, job seekers, and professionals using AI-driven insights. Optimize job applications with resume enhancements and real-time market alignment. Bridge skill gaps through personalized learning paths and promotion road-maps.

4 Problem Statement

NeuraPath solves key problems: career confusion among students due to lack of personalized guidance, high resume rejection rates faced by job seekers from ATS systems, and career stagnation among professionals with**out clear advancement pathways.**

We are developing this project because Existing tools offer fragmented solutions but NeuraPath integrates career mapping, resume optimization and professional growth planning in one platform usin**g cutting-edge AI technologies.**

While some career platforms exist, none offer this complete package. Similar tools either focus only on resume help or generic career advice. NeuraPath uniquely combines all three aspects (student guidance, job search optimization, and professional growth) wit**h live market data integration.**

Building this from scratch will deepen our understanding of AI/ML model deployment, real-time data scraping, and full-stack development. It provides hands- on experience in integra**ting multiple complex systems.**

Through this project, we will master Advanced NLP for resume parsing, Machine learning for career predictions, Web scraping for live job data, Full-stack system architecture.

5 Problem Solution for Proposed System

NeuraPath is an intelligent, all-in-one AI-driven platform designed to address career challenges across different professional stages. For students, it analyzes academic profiles and skills using machine learning models to generate personalized career recommendations while identifying skill gaps. Job seekers benefit from automated resume optimization featuring ATS compliance checks, keyword enhancements, and real-time job market data integration to ensure their applications meet current employer expectations, significantly reducing rejection rates. The platform's AI-powered job matching system connects candidates with relevant opportunities based on their profiles. Professionals gain valuable career progression insights through promotion roadmaps that evaluate their skills, experience, and compensation against industry standards.

Utilizing NLP for resume parsing and web scraping for live job market data, NeuraPath maintains up-to-date recommendations while machine learning models predict career trajectories based on evolving market trends. Its user-friendly interface makes sophisticated career tools accessible to all users, distinguishing itself from fragmented solutions by integrating comprehensive career management in a single platform. Supporting lifelong professional development from initial student choices through NeuraPath eliminates guesswork through its AI-driven approach, delivering accurate, personalized guidance that empowers users to make informed decisions for long-term career success.

6 Related System Analysis /Literature Review

The Systems that already exists:

6.1 LinkedIn Career Explorer

- Provides basic career suggestions but lacks personalized skill gap analysis Offers limited resume optimization features.
- Career recommendations aren't dynamically updated with market trends.
- NeuraPath improves this with AI-driven skill assessment and real-time data integration.

6.2 Myplan Career Assessment

- Uses outdated career databases.
- No integration with current job market data Lacks professional career progression tools.
- Our solution incorporates live labor market analytic and promotion road-maps.

6.3 ResumeWorded

- Focuses only on resume optimization.
- Doesn't connect to career guidance features.
- Limited to English-language resumes.

Table 1: Related System Analysis with Proposed Project Solution

Application Name	Weakness	Proposed Project Solution
LinkedIn Career Explorer	Generic recommendations, no skill gap analysis	AI-powered personalized career mapping with real-time skill assessment
MyPlan Career Assessment	Static career data, no market alignment	Dynamic career predictions using live job market trends
ResumeWorded	Standalone resume tool, no career integration	Unified platform combining resume optimization with career development

7 Advantages/Benefits of Proposed System

- **Personalized Career Guidance -** Uses AI-driven analysis of skills, education, and market trends to provide tailored career recommendations, eliminating guesswork.
- **ATS-Optimized Resumes -** Automatically enhances resumes with job-specific keywords and formatting to bypass automated filters, increasing interview chances.
- **Real-Time Market Alignment -** Integrates live job market data from platforms like LinkedIn and Glassdoor, ensuring recommendations stay current with industry demands.
- **Skill Gap Identification -** Pinpoints missing competencies for desired roles and suggests targeted learning resources (courses, certifications).
- **Career Progression Roadmaps -** Helps professionals plan promotions by analyzing skills, salary benchmarks, and industry growth paths.
- **Unified Platform -** Combines career planning, resume optimization, and job matching in one system, unlike fragmented existing tools.

7.1 Scope

7.1.1 In Scope

- **AI Career Guidance:** ML-based recommendations and skill gap analysis for students.
- **Resume Optimization:** ATS, keyword enhancement, NLP-powered parsing.
- **Job Matching:** Real-time role matching using scraped market data.
- **Career Roadmaps:** Promotion paths and salary benchmarks for professionals.
- **Tech Stack:** NLP (spaCy), Firebase, and web scraping (LinkedIn/Rozee.pk).

7.1.2 Out Scope

- **Direct job application submissions:** The platform will recommend jobs but won't process actual applications.
- **Psychological career assessments:** Career matching is skills-based, excluding personality/psychological testing.

8 Modules

8.1 Module 1: User Onboarding & Smart Profile Management

Handles secure account creation with email verification and role-based access control. Maintains comprehensive user profiles including education, skills, and work experience.

8.2 Module 2: AI Career Intelligence Engine

Uses ML algorithms to analyze profiles and suggest personalized career paths. Integrates real-time labor market data for current industry-aligned recommendations.

8.3 Module 3: Smart Resume Enhancer

Enhances resumes with ATS-compliant formatting and targeted keyword optimization. Generates improvement reports highlighting skill gaps and suggested edits.

8.4 Module 4: AI Job Matcher

Automatically pairs user profiles with suitable job openings using similarity scoring. Continuously updates matches through real-time scraping of major job portals.

8.5 Module 5: Growth Pathway Analyzer

Creates customized promotion roadmaps with skill development timelines. Benchmarks user profiles against industry salary standards and role requirements.

8.6 Module 6: Skill Insight & Learning Advisor

Compares user competencies with target role requirements through AI analysis. Recommends specific courses and certifications to address missing skills.

9 System Limitations/Constraints

- **Data Dependency**

The system's accuracy relies on the quality and freshness of scraped job market data. Inconsistent or incomplete data from external sources (LinkedIn, Glassdoor, etc.) may affect recommendation reliability.

- **Niche Industry Coverage**

Career guidance is optimized for mainstream professions but may lack depth for highly specialized or emerging fields with limited available data.

- **User Input Reliance**

Recommendations depend on accurate self-reported user profiles; incomplete or inaccurate input may reduce system effectiveness.

10 Software Process Methodology

We will use Object-Oriented Methodology (OOM) for developing NeuraPath due to its modularity, reusability, and scalability. This approach aligns perfectly with our system's architecture, which consists of multiple interconnected modules (e.g., Resume Optimizer, Career Recommender) that can be modeled as objects with defined properties and behaviors.

- **Modularity:**

Matches our 9 independent yet interconnected modules.

- **Reusability:**

Common components can be inherited across modules.

- **Scalability:**

New features (e.g., additional career domains) can be added without disrupting existing functionality.

11 Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

	Tools	**Version**	**Rationale**
Tools And Technologies	Android Studio	2022.3.1	IDE
	Firebase	-	Data Storage
	Firebase Authentication	-	User Account Management
	MS Word PowerPoint	2021	Documentation
	Figma	-	UI/UX Design
	Firebase Extensions Mock API	-	APIs
	Cloudinary	-	Media Management
	JUnit/Espresso	4.13.2/3.5.0	Testing
	Technology	**Version**	**Rationale**
	Firebase ML Kit scikit-learn/spaCy	-	AI/ML
	GetX	4.6.5	State Management
	BeautifulSoup/Selenium	4.12+/4.8	Web Scraping
	Git/GitHub	2.40+	Version Control
	Firebase	3.0.0	Backend

12 Project Stakeholders and Roles

Table 3: Project Stakeholders for Proposed Project

Stakeholder	**Role**
COMSATS University	Project Sponsor
Danish Riasat	User Onboarding & Smart Profile Management, Skill Insight & Learning Advisor, Documentation, AI Job Matcher
Usama Abbas	AI Career Intelligence Engine, Growth Pathway Analyzer, Smart Resume Enhancer
Supervisor	Mr. Bilal Shabir Qaisar
FYP Evaluation Committee	Evaluation and Feedback

13 Team Members and Individual Tasks/Work Division

Table 4: Team Member Work division for Proposed Project

Student Name	**Reg. No.**	**Responsibilities**
Danish Riasat	FA22-BCS-219	Developer (Module 1,3,6) + Documentation
Usama Abbas	FA22-BCS-227	Developer (Module 2,4,5)

14 Data Gathering Approach

- To ensure accurate and comprehensive requirements, we will use:
- Surveys & Questionnaires – Distributed to students, job seekers, and professionals to understand their career guidance needs and pain points.
- Interviews – Conducted with university career counselors and HR professionals to gather expert insights on resume screening and career trends.
- Competitor Analysis – Evaluating existing career platforms (e.g., LinkedIn Career explorer, MyPlan) to identify gaps and best practices.
- Web Scraping – Extracting real-time job market data (skills in demand, salary benchmarks) from platforms like LinkedIn and Rozee.pk.
- User Testing – Collecting feedback through prototype testing with target users to refine features like resume optimization and career recommendations.

15 Concepts

15.1 Concept 1: Natural Language Processing (NLP)

Implemented using spaCy/Firebase ML Kit for resume parsing and skill extraction. Enables automated keyword optimization and ATS compliance checks.

15.2 Concept 2: Machine Learning for Career Prediction

Scikit-learn models trained on O*NET datasets to suggest career paths. Involves feature engineering (skills, education) and clustering algorithms.

15.3 Concept 3: Real-Time Data Integration

Web scraping (BeautifulSoup/Selenium) for live job market trends. Firebase synchronization for dynamic content updates.

15.4 Concept 4: Cross-Platform State Management

GetX architecture for Flutter app state handling.

Manages complex user workflows (resume edits → career suggestions).

15.5 Concept 5: Cloud-Native Development

Firebase services (Auth, store) as backend. Cloudinary for media storage/processing.

16 Conclusion

NeuraPath is an AI-driven platform that revolutionizes career guidance by providing personalized recommendations, resume optimization, and skill development roadmaps. Leveraging machine learning and real-time data analysis, it bridges the gap between education and employment needs. The project demonstrates effective integration of cutting-edge technologies to address critical career challenges. By empowering users with data-driven insights, it enhances employment and career decision-making. This implementation showcases the potential of AI in transforming career development services.